

Student Name _____
AM KM



KAMBALA Mathematics Extension 1

HSC Assessment Task 3

June 24, 2008

Time Allowed: 50 minutes

INSTRUCTIONS

- This task contains three questions. Marks for each part of each question are shown.
- Answer all questions in the writing booklets provided. **Start each question in a new booklet.**
- Calculators may be used.
- Show all necessary working.
- Marks may be deducted for careless or badly arranged work.
- More marks will be awarded for questions involving higher order thinking skills.

Kambala Mathematics Extension 1 HSC Task #3, June 24, 2008

Question 1 (start a new booklet)

Marks

- (a) Use the substitution $u = 1 - 2x$ to evaluate

(3)

$$\int_0^1 x(1-2x)^4 dx$$

- (b) Using the substitution $u = e^x$, or otherwise, find

(3)

$$\int_0^{\ln 5} \frac{1}{1+e^{-x}} dx$$

- (c) (i) Express $\sin x - \sqrt{3} \cos x$ in the form $A \sin(x - \alpha)$, with $A > 0$ and $0 < \alpha < \frac{\pi}{2}$.

(2)

- (ii) Find the general solution to $\sin x - \sqrt{3} \cos x = \frac{2}{\sqrt{2}}$.

(2)

Question 2 (start a new booklet)

Marks

- (a) If $\tan \theta = m$ and $\tan \phi = 3$, find the value of m if $\theta - \phi = \frac{\pi}{4}$.

(2)

- (b) Find the equation of the tangent to $y = \tan^{-1}(ax + b)$ at the point where it crosses the x -axis.

(3)

- (c) Sketch the graph of $y = \frac{3}{\pi} \cos^{-1} \frac{x}{2}$ for the domain $-2 \leq x \leq 2$.

(2)

- (d) Find the exact value of $\int_0^3 \frac{1}{\sqrt{12-x^2}} dx$.

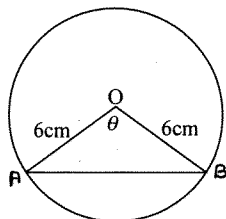
(2)

- (e) What is the condition for the inverse of a function to exist?

(1)

Question 3 (start a new booklet)**Marks**

- (a) O is the centre of a circle of radius 6cm .
 $\angle AOB = \theta$ radians and θ is increasing at a rate of 0.2 radians per second. (3)

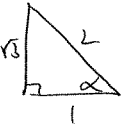
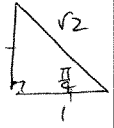


- Describe the rate of change of the area of $\triangle AOB$ when $\angle AOB = 120^\circ$.
- (b) A particle, when x metres from the origin on a straight line, has velocity $v \text{ ms}^{-1}$, given by $v^2 = 20 + 16x - 4x^2$.
- (i) Show that at this point, its acceleration is $-4(x-2) \text{ ms}^{-2}$. (1)
- (ii) State the period of the motion and find its amplitude. (3)
- (c) A particle moves on the x -axis so that, when x metres from the origin, its acceleration is $-2x^3 \text{ ms}^{-2}$. The particle is initially at rest at $x = 4$.
- (i) In what direction will the particle first move, and why? (1)
- (ii) Show that its velocity, $v \text{ ms}^{-1}$, is given by $v^2 = 256 - x^4$. (2)
- (iii) Where will the particle next come to rest and what will be the subsequent motion? (2)

End of assessment task

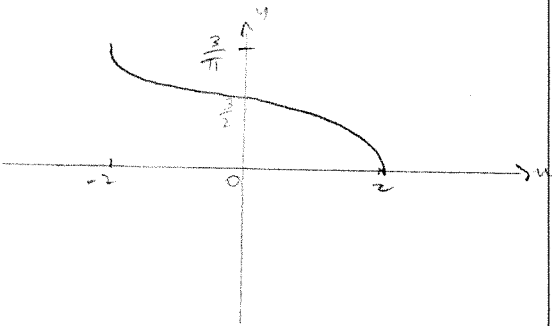
**Year 12 Extension 1 Task 3 June 2008: Solutions**

Qn	Solutions	Marks	Comments+Criteria
1	(a) $\int_0^1 x(1-2x)^4 dx$ $u = 1-2x \rightarrow x = \frac{u-1}{-2}$ $du = -2 dx$ $x=1, u=1-2=-1$ $x=0, u=1-0=1$ $I = -\frac{1}{2} \cdot 2 \int_1^{-1} (u-1)u^4 du$ $= -\int_1^{-1} u^5 - u^4 du$ $= \left[\frac{u^6}{6} - \frac{u^5}{5} \right]_1^{-1}$ $= \left(\frac{1}{6} + \frac{1}{5} \right) - \left(\frac{1}{6} - \frac{1}{5} \right)$ $= \frac{2}{5}$	1	
	(b) $\int_0^5 \frac{1}{1+e^x} dx$; $u = e^x \rightarrow \frac{du}{dx} = e^x \rightarrow du = e^x dx$ $x = \ln 5, u = e^{\ln 5} = 5$ $x=0, u = e^0 = 1$ $I = \int_1^5 \frac{1}{1+u} \cdot \frac{du}{u}$ $= \int_1^5 \frac{1}{u+1} du$ $= \left[\ln(u+1) \right]_1^5$ $= \ln 6 - \ln 2$	1	$u^{-1} = (e^x)^{-1} = e^{-x}$ $dx = \frac{du}{e^x}$ $= \frac{du}{u}$

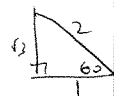
Qn	Solutions	Marks	Comments+Criteria
1	(c) (i) $\sin x - \sqrt{3} \cos x \equiv A \sin(x - \alpha)$ $A = \sqrt{a^2 + b^2} \quad a = 1, b = \sqrt{3}$ $= \sqrt{1+3}$ $= \sqrt{4}$ $= 2$ $\tan \alpha = \frac{b}{a} = \frac{\sqrt{3}}{1}$  $\alpha = \frac{\pi}{3}$ $\therefore \sin x - \sqrt{3} \cos x \equiv 2 \sin(x - \frac{\pi}{3})$		
	(ii) $\sin x - \sqrt{3} \cos x = \frac{2}{\sqrt{2}}$ (S) (A) $2 \sin(x - \frac{\pi}{3}) = \frac{2}{\sqrt{2}}$ $\sin(x - \frac{\pi}{3}) = \frac{1}{\sqrt{2}}$  basic angle, $x - \frac{\pi}{3} = \frac{\pi}{4}$ $\theta = n\pi \pm (-1)^n \alpha$ $x - \frac{\pi}{3} = n\pi \pm (-1)^n \frac{\pi}{4}$ $x = n\pi \pm (-1)^n \frac{\pi}{4} + \frac{\pi}{3}$ i) if n odd, $x = n\pi - \frac{\pi}{4} + \frac{\pi}{3}$ $= n\pi + \frac{\pi}{12}$ if n even or 0, $x = n\pi + \frac{\pi}{4} + \frac{\pi}{3}$ $x = n\pi + \frac{7\pi}{12}$		

Qn	Solutions	Marks	Comments+Criteria
2	(a) $\tan \theta = m, \tan \phi = 3$ $\theta - \phi = \frac{\pi}{4}$ $\tan(\theta - \phi) = \tan \frac{\pi}{4}$ $\therefore \frac{\tan \theta - \tan \phi}{1 + \tan \theta \tan \phi} = 1$ i) $\frac{m - 3}{1 + 3m} = 1$ $m - 3 = 1 + 3m$ $-2m = 4$ $m = -2$		
	(b) $y = \tan^{-1}(ax + b)$ $\frac{dy}{dx} = \frac{1}{1 + (ax + b)^2} \cdot a$ $= \frac{a}{1 + (ax + b)^2}$ when $y = 0, \tan^{-1}(ax + b) = 0$ $\tan 0 = ax + b$ $\frac{dy}{dx} = \frac{a}{1 + (\tan 0)^2}$ $= a$ $y - 0 = a(x - (ax + b))$ $= a(x - ax - b)$ $y = ax - a^2x - ab$		

Year 12 Extension 1 Task 3 June 2008: Solutions

Qn	Solutions	Marks	Comments+Criteria
2 (c)	$y = \frac{3}{\pi} \cos^{-1} \frac{x}{2}$ for $-2 \leq x \leq 2$ 		
(d)	$\int_0^3 \frac{1}{\sqrt{12-x^2}} dx \quad \sqrt{12} = 2\sqrt{3}$ $= \int_0^3 \frac{1}{\sqrt{(2\sqrt{3})^2 - x^2}} dx$ $= \left[\sin^{-1} \left(\frac{x}{2\sqrt{3}} \right) \right]_0^3$ $= \sin^{-1} \left(\frac{3}{2\sqrt{3}} \right) - \sin^{-1} 0$ $= \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) - \sin^{-1} 0$ $= \frac{\pi}{3}$		
(e)	if $y=f(x)$, then $y=f^{-1}(y)$ exists iff for every x value there exists one and only one value for y.		

Year 12 Extension 1 Task 3 June 2008: Solutions

Qn	Solutions	Marks	Comments+Criteria
3 (a)	$\frac{dr}{dt} = 0.2 \text{ m/sec}$ $\frac{dA}{dt} = ? \text{ when } \theta = 120^\circ$ $A = \frac{1}{2} r^2 \sin \theta$ $\frac{dA}{dr} = r \sin \theta$ $\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$ $= r \sin \theta \cdot \frac{dr}{dt}$ $= 6 \sin 120^\circ \cdot 0.2$ $= 6 \cdot \frac{\sqrt{3}}{2} \cdot 0.2$ $= 3\sqrt{3} \cdot 0.2$ $= \frac{3\sqrt{3}}{5}$ 		
(b)	$v^2 = 20 + 16x - 4x^2$ $(i) \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d}{dx} (10 + 8x - 2x^2)$ $= 8 - 4x$ $= -4(x-2) \text{ m/s}^2$		
(ii)	$a = -4(x-2)$ $a^2 = 4$ $a = 2$ $\therefore \text{period} = \frac{2\pi}{a} = \pi$		

Year 12 Extension 1 Task 3 June 2008: Solutions

Qn	Solutions	Marks	Comments+Criteria
3	(c) $a = -2x^3$ m/s² $t=0, x=4$ (i) Initially, the particle will move left of $x=4$ (in a neg. direction) since $a = -2 < 0$.		
	(ii) $v^2 = 256 - x^4$ $a = -2x^3$ $v = \frac{-2x^4}{4} + C$ $= -\frac{x^4}{2} + C$ when $t=0, v=0, x=4$. $0 = -\frac{4^4}{2} + C$ $= -128 + C$ $\therefore C = 128$ $v = -\frac{x^4}{2} + 128 \rightarrow v^2 = \left(-\frac{x^4}{2} + 128\right)^2$ $v^2 = \frac{1}{4}(x^4 - 256)^2$ $= \frac{1}{4}(x^8 - 512x^4 + 256^2)$ $= \frac{1}{4}x^8 - 128x^4 + 16384$		$a = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$ $= \frac{d}{dx}\left(\frac{1}{2}(256 - x^4)\right)$ $= \frac{d}{dx}\left(128 - \frac{x^4}{2}\right)$ $= -\frac{4x^3}{2}$ $= -2x^3$

Year 12 Extension 1 Task 3 June 2008: Solutions

Qn	Solutions	Marks	Comments+Criteria
3	(c) and (iii) $v^2 = 256 - x^4$ $v = \sqrt{256 - x^4}$ when $v=0, 256 - x^4 = 0$ $x^4 = 256$ $x = \pm 4$ $\therefore$ particle at rest initially at $x=4$ and again at $x=-4$. at $x=-4, a = -2(-4)^3$ $= -2 \times -64$ $= 128 > 0$ $\therefore$ particle is moving in neg. dirn		